

Chapter 5 (cont): Randomized Complete Block Designs

RCBD

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STAT452

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CRD Review

Review: the one-way CRD model

With g treatments and n_i replicates per treatment:

$$y_{ij} = \mu_i + \varepsilon_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma^2).$$

- ▶ All “unexplained” variation is dumped into the **error** ε_{ij} .
- ▶ If the experimental units differ a lot $\Rightarrow$ large $\sigma^2 \Rightarrow$ large MSE $\Rightarrow$ a **low-power** F -test.

Question

If we *know in advance* of a large source of variation (e.g. store, batch of raw material, time period, patient. . .), should we leave it sitting in the error term or should we **model** it and remove it?

Example 1 — Air Freshener Sales

Goal: compare 4 treatments (price + display) on the sales of an air-freshener brand.

A high price + extra display

B medium price + extra display

C low price + extra display

D medium price + **no** extra display

- ▶ Run in **8 stores**, each store over **4 weeks**, applying *one* treatment per week.
- ▶ Response: number of units sold in a week.
- ▶ **Blocking idea:** the 8 stores differ greatly in size/clientele $\Rightarrow$ take the *store* as a block.

Example 1 — Raw data

Week	Store							
	1	2	3	4	5	6	7	8
1	B 31	A 23	C 12	D 3	A 10	C 30	B 23	D 14
2	A 19	D 16	B 14	C 4	B 21	D 25	C 17	A 14
3	D 15	C 30	A 12	B 6	C 12	A 47	D 5	B 3
4	C 16	B 27	D 5	A 11	D 12	B 38	A 13	C 6

- ▶ Each cell records a *treatment* and its corresponding *sales*.
- ▶ Within each store, all 4 treatments appear exactly **once**.
- ▶ The treatment-to-week order is **randomized** separately for each store.

What if we analyse it as a CRD (ignoring stores)?

Treat the data as a one-way CRD with 4 treatments, $y_{ij} = \mu_i + \varepsilon_{ij}$:

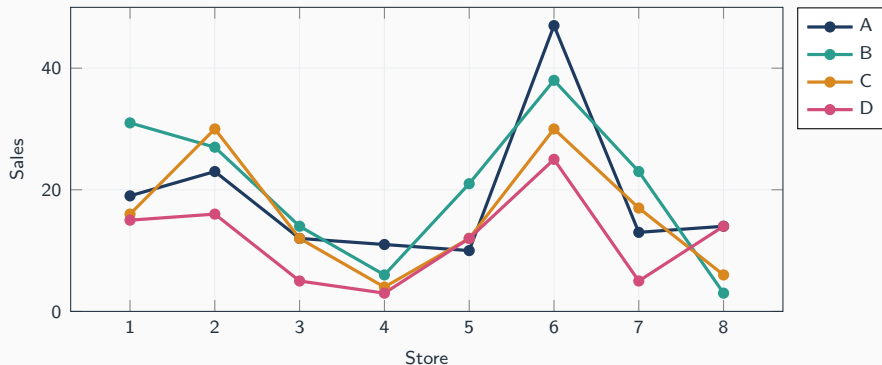
```
Response: sales
          Df Sum Sq Mean Sq F value Pr(>F)
trtmt      3    329   109.8    0.984  0.414
Residuals 28   3123   111.5
```

- ▶ $F = 0.98$, $p = 0.41 \Rightarrow$ **no** evidence of a treatment difference (!).
- ▶ But $\text{MSE} = 111.5$ is huge — most of it is *store-to-store variation* locked inside the error.

Do not conclude too fast

This “non-significant” result may be **spurious**: store noise is masking the treatment signal. Let us look carefully at the data by store.

Why must we account for the store effect?



- Store-to-store variation is large (store 6 is far above, store 4 far below).
 - Within *each* store, treatment D is almost always the lowest $\Rightarrow$ there is a sign of a treatment effect.
- $\Rightarrow$ We should put “store” into the model as a **block**.

The Blocking Principle

What is a block?

Definition — Block

A **block** is a set of experimental units that is *homogeneous* in some sense. Expectation: units within the same block will respond *similarly* if given the same treatment.

Definition — Block design

Randomize the assignment of units to treatments *within each block*. The randomization is *restricted*: a unit never “leaves” its block.

- Stores, patients, batches of material, field plots, production days... can all serve as blocks.

Things that can be used as blocks

- ▶ When a known **source of variation** can be identified:
 - ▶ subject characteristics: age, sex, medical history, baseline weight;
 - ▶ machine / technician / production lot (e.g. milk collected in one day).
- ▶ **Spatial** blocking: nearby field plots, position within a furnace.
- ▶ **Temporal** blocking: day, shift, measurement batch.
- ▶ **Matched-pair** blocking: twins, the two eyes of one person, before–after on the same subject.

Remark

A good block makes *units within it similar* and makes *blocks differ a lot from one another*. It is precisely the *between-block* difference that we want to “pull” out of the error.

The benefit of blocking

The second principle of experimental design

“Block what you can, randomize everything else.”

- ▶ Blocking is the **second** fundamental principle, right after **randomization**.
- ▶ If units are highly variable, grouping them into similar blocks can **sharply reduce noise** $\Rightarrow$ **increase the power** to detect treatment differences.
- ▶ **Choosing the block is critical**: a poor choice (blocks that are not internally homogeneous) yields no gain and may even cost degrees of freedom.

Randomizing within blocks — an example

Suppose $g = 4$ treatments, and a block is one store run over 4 weeks. We randomly assign the treatment order to the 4 weeks *independently* for each store:

	Week 1	Week 2	Week 3	Week 4
Store 1	B	A	D	C
Store 2	A	D	C	B
Store 3	C	B	A	D
		⋮		

- ▶ Each block contains **all** 4 treatments (hence “**complete**”).
- ▶ The order permutation is **random** and **separate** for each block.
- ▶ In R: `sample(c("A", "B", "C", "D"))` for each block.

Structure of the RCBD

Formal definition of the RCBD

There are g treatments to compare and b available blocks of units; each block contains $k = rg$ units.

- ▶ Within each block, the $k = rg$ units are **randomized** to the g treatments, each treatment receiving r units.
- ▶ “**Complete**”: each treatment appears the *same number of times* (r) in *every* block.
- ▶ Most common: block size $k = g$, i.e. $r = 1$ (each treatment exactly once per block). This is the **RCBD** in the narrow sense — our focus today.
- ▶ A **matched-pair** design is a special RCBD with block size $k = 2$.

Terminology

If $k > g$ (each treatment replicated $r > 1$ times per block) → **generalized complete block design**. If $k < g$ (a block cannot hold all treatments) → **incomplete block design** — a later chapter.

Laying out RCBD data ($r = 1$)

	Block 1	Block 2	...	Block b
Treatment 1	y_{11}	y_{12}	...	y_{1b}
Treatment 2	y_{21}	y_{22}	...	y_{2b}
$\vdots$	$\vdots$	$\vdots$		$\vdots$
Treatment g	y_{g1}	y_{g2}	...	y_{gb}

- ▶ A total of $N = gb$ observations; this is a **complete, balanced** table.
- ▶ Data are usually arranged by block $\times$ treatment.

Remark

You **cannot** tell from the data table what was or was not randomized. The design lives in *how the experiment was carried out*, not in the resulting numbers.

Mean notation — the dot convention

For balanced RCBD data ($r = 1$), we use a $\bullet$ to indicate the index averaged over:

$$\bar{y}_{i\bullet} = \frac{1}{b} \sum_{j=1}^b y_{ij} \text{ mean of treatment } i \text{ (averaged over blocks).}$$

$$\bar{y}_{\bullet j} = \frac{1}{g} \sum_{i=1}^g y_{ij} \text{ mean of block } j \text{ (averaged over treatments).}$$

$$\bar{y}_{\bullet\bullet} = \frac{1}{gb} \sum_{i=1}^g \sum_{j=1}^b y_{ij} \text{ the grand mean.}$$

Remark

These three quantities are the “building blocks” for every estimate and sum of squares to come. Remember: a $\bullet$ means “averaged over that index”.

The RCBD Model & Identifiability Constraints

The model for an RCBD

Definition — RCBD model (additive)

$$y_{ij} = \mu + \underbrace{\alpha_i}_{(\text{trt})} + \underbrace{\beta_j}_{(\text{block})} + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma^2).$$

where

y_{ij} response of the unit given treatment i in block j ;

μ the grand mean;

α_i the **treatment effect**, $i = 1, \dots, g$;

β_j the **block effect**, $j = 1, \dots, b$;

ε_{ij} measurement error, i.i.d. $N(0, \sigma^2)$.

Question

In an RCBD, do we care more about α_i or β_j ? $\Rightarrow \alpha_i$ (treatments). The block is only a noise-control device.

The model is over-parameterized — why constraints?

The additive model has $1 + g + b$ location parameters: $\mu, \alpha_1, \dots, \alpha_g, \beta_1, \dots, \beta_b$.

- ▶ But the data can only “see” the *expected cell means* $\mathbb{E}(y_{ij}) = \mu + \alpha_i + \beta_j$.
- ▶ The number of independent degrees of freedom in the mean structure is only

$$\underbrace{1}_{\mu} + \underbrace{(g-1)}_{\alpha} + \underbrace{(b-1)}_{\beta} = g + b - 1.$$

- ▶ We have $1 + g + b$ parameters but only $g + b - 1$ independent constraints $\Rightarrow$ 2 dimensions of slack $\Rightarrow$ we need 2 constraints.

The essence of “over-parameterized”

If you add a constant c to every α_i and subtract c from μ , the model is *unchanged*. The same holds for β_j . The parameters are currently **not identifiable** — constraints must be imposed to pin down a unique solution.

Zero-sum constraints

The standard choice (as in balanced factorial models) is to impose:

$$\sum_{i=1}^g \alpha_i = 0 \quad \text{and} \quad \sum_{j=1}^b \beta_j = 0.$$

These are exactly the 2 missing constraint equations.

- ▶ Then μ is the true overall mean;
- ▶ α_i measures the deviation of treatment i from the grand mean;
- ▶ β_j measures the deviation of block j from the grand mean.

Interpretation

Under this constraint, $\mu + \alpha_i$ is exactly the *mean of treatment i* (averaged over blocks), and $\mu + \beta_j$ is the *mean of block j* .

An alternative: reference (corner) coding

R does *not* use zero-sum constraints by default; it uses **corner / reference constraints (treatment contrasts)**:

$$\alpha_1 = 0 \quad \text{and} \quad \beta_1 = 0$$

(the first level is the “baseline”). Then:

- ▶ μ = expected mean of the cell (treatment 1, block 1);
- ▶ α_i = difference of treatment i *relative to* treatment 1;
- ▶ β_j = difference of block j *relative to* block 1.

Caution when reading R output

The *individual parameter estimates* (`summary(lm)`) depend on the constraint, but the **invariant** quantities — SS_{trt} , SS_{block} , SSE, the F statistic, the p -value, and the **contrasts** $\alpha_i - \alpha_{i'}$ — are **unchanged** whichever constraint is chosen.

Parameter Estimation (Least Squares)

The least-squares problem

Estimate μ, α_i, β_j by minimizing the error sum of squares

$$Q(\mu, \alpha, \beta) = \sum_{i=1}^g \sum_{j=1}^b (y_{ij} - \mu - \alpha_i - \beta_j)^2$$

subject to $\sum_i \alpha_i = 0, \sum_j \beta_j = 0$.

- ▶ This is a constrained optimization $\Rightarrow$ take partial derivatives (the **normal equations**) and impose the constraints.
- ▶ We will obtain very clean **orthogonal, separable** formulas — a consequence of the design being **balanced**.

Derivation: the normal equation for μ

Differentiate Q with respect to μ and set it to 0:

$$\frac{\partial Q}{\partial \mu} = -2 \sum_{i=1}^g \sum_{j=1}^b (y_{ij} - \mu - \alpha_i - \beta_j) = 0.$$

Expand the sum:

$$\sum_i \sum_j y_{ij} = gb\mu + b \sum_i \alpha_i + g \sum_j \beta_j.$$

Impose $\sum_i \alpha_i = 0$, $\sum_j \beta_j = 0$:

$$y_{\bullet\bullet} = gb\hat{\mu} \implies \boxed{\hat{\mu} = \bar{y}_{\bullet\bullet}}$$

Remark

It is precisely the **constraints** that made the two sums $\sum \alpha_i, \sum \beta_j$ vanish, giving $\hat{\mu}$ equal to the grand mean. Without constraints the system has infinitely many solutions.

Derivation: the treatment-effect estimate α_i

Differentiate Q with respect to α_i (fixed i) and set to 0:

$$\frac{\partial Q}{\partial \alpha_i} = -2 \sum_{j=1}^b (y_{ij} - \mu - \alpha_i - \beta_j) = 0.$$

Expand:

$$\sum_j y_{ij} = b\mu + b\alpha_i + \sum_j \beta_j \quad \sum \beta_j = 0 \quad b\mu + b\alpha_i.$$

Divide by b and substitute $\hat{\mu} = \bar{y}_{\bullet\bullet}$:

$$\bar{y}_{i\bullet} = \hat{\mu} + \hat{\alpha}_i = \bar{y}_{\bullet\bullet} + \hat{\alpha}_i \quad \Longrightarrow \quad \boxed{\hat{\alpha}_i = \bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet}} \quad (i = 1, \dots, g).$$

Similarly, differentiating with respect to β_j gives:

$$\boxed{\hat{\beta}_j = \bar{y}_{\bullet j} - \bar{y}_{\bullet\bullet}} \quad (j = 1, \dots, b).$$

Check: the estimates satisfy the constraints

The estimates *automatically* satisfy the zero-sum constraints:

$$\sum_{i=1}^g \hat{\alpha}_i = \sum_i (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet}) = \left(\sum_i \bar{y}_{i\bullet} \right) - g \bar{y}_{\bullet\bullet} = g \bar{y}_{\bullet\bullet} - g \bar{y}_{\bullet\bullet} = 0,$$

since $\frac{1}{g} \sum_i \bar{y}_{i\bullet} = \bar{y}_{\bullet\bullet}$. Entirely similarly, $\sum_j \hat{\beta}_j = 0$.

Theorem — LS estimates for a balanced RCBD

$$\hat{\mu} = \bar{y}_{\bullet\bullet}, \quad \hat{\alpha}_i = \bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet}, \quad \hat{\beta}_j = \bar{y}_{\bullet j} - \bar{y}_{\bullet\bullet}.$$

These coincide with the formulas of the **balanced two-way additive model** — because it is the same orthogonal design structure.

Fitted values and residuals

Fitted values:

$$\hat{y}_{ij} = \hat{\mu} + \hat{\alpha}_i + \hat{\beta}_j = \bar{y}_{\bullet\bullet} + (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet}) + (\bar{y}_{\bullet j} - \bar{y}_{\bullet\bullet}) = \bar{y}_{i\bullet} + \bar{y}_{\bullet j} - \bar{y}_{\bullet\bullet}.$$

Residuals:

$$e_{ij} = y_{ij} - \hat{y}_{ij} = y_{ij} - \bar{y}_{i\bullet} - \bar{y}_{\bullet j} + \bar{y}_{\bullet\bullet}.$$

Remark — why a balanced design is so “clean”

Thanks to balance, the design matrix is *orthogonal*: the estimate $\hat{\alpha}_i$ does **not** depend on $\hat{\beta}_j$ and vice versa. Adding or removing blocks from the model does *not* change the treatment-effect estimates — which is *not* true when the data are unbalanced.

Sum-of-Squares Decomposition

The fundamental decomposition identity

For each observation, write the deviation from the grand mean as three parts:

$$\underbrace{y_{ij} - \bar{y}_{\bullet\bullet}}_{\text{total}} = \underbrace{(\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})}_{\text{treatment}} + \underbrace{(\bar{y}_{\bullet j} - \bar{y}_{\bullet\bullet})}_{\text{block}} + \underbrace{(y_{ij} - \bar{y}_{i\bullet} - \bar{y}_{\bullet j} + \bar{y}_{\bullet\bullet})}_{\text{residual}}.$$

Square and sum over i, j :

$$\text{SST} = \text{SS}_{\text{trt}} + \text{SS}_{\text{block}} + \text{SSE}.$$

What must be shown

Squaring the sum of three terms produces three **cross-product terms**. We must show all three **equal 0** for the identity to hold. This is a consequence of orthogonality.

Proof: the cross terms vanish (1/2)

Let $A_{ij} = \bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet}$, $B_{ij} = \bar{y}_{\bullet j} - \bar{y}_{\bullet\bullet}$, $E_{ij} = y_{ij} - \bar{y}_{i\bullet} - \bar{y}_{\bullet j} + \bar{y}_{\bullet\bullet}$.

Cross term A-B:

$$\sum_i \sum_j A_{ij} B_{ij} = \sum_i \sum_j (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})(\bar{y}_{\bullet j} - \bar{y}_{\bullet\bullet}).$$

Separate the sums (each factor depends on its own index):

$$= \underbrace{\sum_i (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})}_{=0} \cdot \underbrace{\sum_j (\bar{y}_{\bullet j} - \bar{y}_{\bullet\bullet})}_{=0} = 0,$$

since $\sum_i (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet}) = g\bar{y}_{\bullet\bullet} - g\bar{y}_{\bullet\bullet} = 0$ and similarly for blocks.

Proof: the cross terms vanish (2/2)

Cross term A-E:

$$\sum_i \sum_j (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet}) E_{ij} = \sum_i (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet}) \underbrace{\sum_j (y_{ij} - \bar{y}_{i\bullet} - \bar{y}_{\bullet j} + \bar{y}_{\bullet\bullet})}_{(*)}.$$

Consider (*): $\sum_j y_{ij} = b\bar{y}_{i\bullet}$, $\sum_j \bar{y}_{i\bullet} = b\bar{y}_{i\bullet}$, $\sum_j \bar{y}_{\bullet j} = b\bar{y}_{\bullet\bullet}$, $\sum_j \bar{y}_{\bullet\bullet} = b\bar{y}_{\bullet\bullet}$, so

$$(*) = b\bar{y}_{i\bullet} - b\bar{y}_{i\bullet} - b\bar{y}_{\bullet\bullet} + b\bar{y}_{\bullet\bullet} = 0 \Rightarrow \text{cross term A-E} = 0.$$

Similarly, **cross term B-E** = 0 (swap the roles of the sum over i).

Theorem — Orthogonal decomposition

$$\sum_{i,j} (y_{ij} - \bar{y}_{\bullet\bullet})^2 = \text{SS}_{\text{trt}} + \text{SS}_{\text{block}} + \text{SSE},$$

with

$$\text{SS}_{\text{trt}} = b \sum_i (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})^2, \quad \text{SS}_{\text{block}} = g \sum_j (\bar{y}_{\bullet j} - \bar{y}_{\bullet\bullet})^2, \quad \text{SSE} = \sum_{i,j} (y_{ij} - \bar{y}_{i\bullet} - \bar{y}_{\bullet j} + \bar{y}_{\bullet\bullet})^2.$$

Degrees of freedom

Source	Total	Treatment	Block	Error
Symbol	SST	SS_{trt}	SS_{block}	SSE
d.f.	$gb - 1$	$g - 1$	$b - 1$	$(g - 1)(b - 1)$

Check that the degrees of freedom add up:

$$(g - 1) + (b - 1) + (g - 1)(b - 1) = (g - 1) + (b - 1) + (gb - g - b + 1) = gb - 1. \checkmark$$

Remark on the “price” of blocking

Compared to a CRD with the same $N = gb$ units (error d.f. = $gb - g = g(b - 1)$), the RCBD “**borrows**” $b - 1$ degrees of freedom for the block. We *lose* $b - 1$ error d.f. but *gain* the removal of block variation from the noise. Worth it when the blocks really do differ.

Mean squares

Mean square = sum of squares divided by degrees of freedom:

$$MS_{\text{trt}} = \frac{SS_{\text{trt}}}{g - 1}, \quad MS_{\text{block}} = \frac{SS_{\text{block}}}{b - 1}, \quad MSE = \frac{SSE}{(g - 1)(b - 1)}.$$

Theorem — Expected mean squares (EMS)

Under the RCBD model with constraints $\sum \alpha_i = \sum \beta_j = 0$:

$$\mathbb{E}(MS_{\text{trt}}) = \sigma^2 + \frac{b}{g - 1} \sum_{i=1}^g \alpha_i^2, \quad \mathbb{E}(MS_{\text{block}}) = \sigma^2 + \frac{g}{b - 1} \sum_{j=1}^b \beta_j^2,$$

$$\mathbb{E}(MSE) = \sigma^2.$$

$\Rightarrow$ MSE is an unbiased estimator of σ^2 .

Sketch of the derivation of $\mathbb{E}(\text{MSE}) = \sigma^2$

The residual is $e_{ij} = y_{ij} - \bar{y}_{i\bullet} - \bar{y}_{\bullet j} + \bar{y}_{\bullet\bullet}$. Substituting $y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}$:

$$e_{ij} = \varepsilon_{ij} - \bar{\varepsilon}_{i\bullet} - \bar{\varepsilon}_{\bullet j} + \bar{\varepsilon}_{\bullet\bullet}$$

(every μ, α_i, β_j component **cancel**s — this is why the residual is “clean” of the fixed effects).

- ▶ $\text{SSE} = \sum_{i,j} e_{ij}^2$ is a quadratic form in the ε_{ij} .
- ▶ One can show $\mathbb{E}(\text{SSE}) = (g-1)(b-1)\sigma^2$ (the degrees of freedom times σ^2).
- ▶ Hence $\mathbb{E}(\text{MSE}) = \frac{\mathbb{E}(\text{SSE})}{(g-1)(b-1)} = \sigma^2$.

Meaning

When H_0 (all $\alpha_i = 0$) holds, $\mathbb{E}(\text{MS}_{\text{trt}}) = \sigma^2 = \mathbb{E}(\text{MSE})$: the numerator and denominator of F both estimate σ^2 , so $F \approx 1$. When H_0 is false, the numerator inflates $\Rightarrow F$ is large.

The F-test & the ANOVA Table

The F-test for the treatment effect

The hypothesis of equal treatment effects:

$$H_0 : \alpha_1 = \alpha_2 = \cdots = \alpha_g \quad \text{vs.} \quad H_a : \text{not all } \alpha_i \text{ are equal.}$$

Because of the constraint $\sum_i \alpha_i = 0$, this is equivalent to

$$H_0 : \alpha_1 = \cdots = \alpha_g = 0 \quad \text{vs.} \quad H_a : \text{at least one } \alpha_i \neq 0.$$

The test statistic:

$$F_{\text{trt}} = \frac{\text{MS}_{\text{trt}}}{\text{MSE}} \sim F_{g-1, (g-1)(b-1)} \quad \text{under } H_0.$$

Reject H_0 when F_{trt} is large (small p -value).

The ANOVA table for an RCBD

Source	d.f.	SS	MS	F
Block	$b - 1$	SS_{block}	MS_{block}	$(F_{\text{block}} = \frac{MS_{\text{block}}}{MSE})$
Treatment	$g - 1$	SS_{trt}	MS_{trt}	$F_{\text{trt}} = \frac{MS_{\text{trt}}}{MSE}$
Error	$(g - 1)(b - 1)$	SSE	MSE	
Total	$gb - 1$	SST		

Caution — do not test the block

F_{block} is usually **parenthesized / dropped**. The deeper reason: the block is **not** randomized (a unit is always inside its block), so an “ F -test for the block” does not correspond to a valid randomization test. We block because we *believe* blocks differ — rediscovering that carries no inferential meaning.

Why not include a block×treatment interaction?

- ▶ With $r = 1$ (one observation per cell), a full model with interaction $(\alpha\beta)_{ij}$ would use **all** $gb - 1$ degrees of freedom $\Rightarrow$ **none left** for error, and σ^2 cannot be estimated.
- ▶ We **assume additivity**: block and treatment do not interact. The SSE term is precisely the “container” for any interaction — it is used as the surrogate error.
- ▶ Inferentially: we want to draw conclusions *generalizing* over all blocks, so this is the *correct* error against which to test the treatment effect.

Remark

The additivity assumption is *forced* by the lack of degrees of freedom, not always true. If a multiplicative interaction is suspected, one can use the **Tukey one-degree-of-freedom test** (later) or transform the response scale.

The Efficiency of Blocking

Comparing ANOVA tables: RCBD vs. CRD

If we **ignore** the block effect and analyse the same data as a CRD:

Source	d.f.	SS	MS	F
Treatment	$g - 1$	SS_{trt}	MS_{trt}	$F = \frac{MS_{\text{trt}}}{MSE_{\text{CRD}}}$
Error	$gb - g$	SSE_{CRD}	MSE_{CRD}	
Total	$gb - 1$	SST		

Both tables have the **same** SS_{trt} , but the block variation now sits inside the error:

$$SSE_{\text{CRD}} = SSE_{\text{RCBD}} + SS_{\text{block}}$$

Consequence

If SS_{block} is large, putting the block into the model **sharply reduces noise** $\Rightarrow$ it is easier to detect treatment differences.

Relative Efficiency (RE)

Definition — RE of the RCBD relative to the CRD

RE is the factor by which a CRD's sample size would have to be **multiplied** to obtain the same amount of information (the same precision) as the current RCBD.

Estimated error variance of the hypothetical CRD (a weighted average):

$$\hat{\sigma}_{\text{CRD}}^2 = \frac{(b-1)\text{MS}_{\text{block}} + b(g-1)\text{MSE}}{gb-1}, \quad \hat{\sigma}_{\text{RCBD}}^2 = \text{MSE}.$$

The RE formula with the (Fisher) **degrees-of-freedom correction**:

$$\text{RE} = \frac{(\nu_r + 1)(\nu_c + 3)}{(\nu_r + 3)(\nu_c + 1)} \cdot \frac{\hat{\sigma}_{\text{CRD}}^2}{\hat{\sigma}_{\text{RCBD}}^2},$$

with $\nu_r = (g-1)(b-1)$, $\nu_c = g(b-1)$.

Interpreting and using RE

- ▶ The first factor is the **degrees-of-freedom correction** (usually near 1): the variance is estimated more precisely with more degrees of freedom.
- ▶ The second factor — the **variance ratio** — mostly determines the size of RE.
- ▶ $RE > 1$: blocking *pays off*. $RE = 3$ means a CRD would need **three times** as many units to achieve the same precision.
- ▶ Units a CRD would need $\approx RE \times (gb)$.

When does blocking backfire?

If the blocks in fact do *not* differ ($MS_{\text{block}} \approx MSE$), we still lose $b - 1$ error degrees of freedom without reducing noise $\Rightarrow RE < 1$, a weaker test than the CRD. Hence **block choice must be well founded**.

Worked RE example — the Auditor Training data

From the RCBD ANOVA table (computed in the R section): $b = 10$ blocks, $g = 3$ treatments,

$$MS_{\text{block}} = \frac{433.4}{9} = 48.16, \quad MSE = \frac{112.3}{18} = 6.24.$$

Step 1 — hypothetical CRD variance:

$$\hat{\sigma}_{\text{CRD}}^2 = \frac{(10 - 1)(48.16) + 10(3 - 1)(6.24)}{10 \cdot 3 - 1} = \frac{433.4 + 124.8}{29} = 19.25.$$

Step 2 — variance ratio: $\frac{19.25}{6.24} = 3.09.$

Step 3 — d.f. correction ($\nu_r = 18$, $\nu_c = 27$):

$$\frac{(18 + 1)(27 + 3)}{(18 + 3)(27 + 1)} = \frac{19 \cdot 30}{21 \cdot 28} = 0.969.$$

$$RE = 0.969 \times 3.09 \approx 3.0$$

Interpreting the Auditor Training result

Remark

$RE \approx 3.0$: by blocking on *time of graduation*, the experiment is **three times** as efficient. A CRD wanting the same precision would need $\approx 3 \times 30 = 90$ auditors instead of 30.

A quick check from the raw sums of squares:

- ▶ RCBD: $MSE = 6.24$ (d.f. 18).
- ▶ CRD (ignoring blocks): $SSE_{CRD} = 112.3 + 433.4 = 545.7$, d.f. 27,
 $MSE_{CRD} = 545.7/27 = 20.2$.
- ▶ The ratio $20.2/6.24 \approx 3.2$ — the same message: noise cut roughly **threefold**.

(For the Air Freshener data: similarly, $RE \approx 3.3$ — blocking on store is well worth it.)

Additivity & Multiple Comparisons

The additivity assumption and how to check it

The RCBD model assumes block and treatment are **additive**: the difference between treatments is the *same* in every block.

Caution

Additivity is not automatically true. If a *multiplicative* interaction exists (transformable nonadditivity), the residuals inflate and the test loses validity.

Tukey one-degree-of-freedom test for nonadditivity:

1. Fit the additive model; obtain the fitted values $\hat{y}_{ij}$.
2. Create a new predictor $z_{ij} = \hat{y}_{ij}^2 / (2\bar{y})$.
3. Fit the model augmented with z_{ij} ; the extra SS is SS_{Tukey} (1 d.f.).
4. $F = \frac{SS_{\text{Tukey}}/1}{SSE^* / [(g-1)(b-1) - 1]}$; a large $F \Rightarrow$ nonadditivity is present.

What if nonadditivity is detected?

- ▶ The Tukey interaction form $(\alpha\beta)_{ij} = \eta \alpha_i \beta_j / \mu$ can often be *removed* by a power transformation of the response with exponent $1 - \eta$.
- ▶ Tool for choosing the transformation: **Box–Cox** (pick λ by maximizing the likelihood).
- ▶ On the “right” scale the data are more additive $\Rightarrow$ SSE contains less interaction $\Rightarrow$ a better surrogate error.

Practical remark

A sensible order: (1) plot the block $\times$ treatment interaction; (2) run the Tukey test; (3) if needed, transform the response; (4) re-analyse. Beware: a transformation that reduces interaction can sometimes make the variance heterogeneous.

Multiple comparisons after ANOVA

Once H_0 is rejected, we compare treatments pairwise. The standard error of the difference between two treatment means (each based on b blocks):

$$\text{SE}(\bar{y}_{i\bullet} - \bar{y}_{i'\bullet}) = \sqrt{\text{MSE} \left(\frac{1}{b} + \frac{1}{b} \right)} = \sqrt{\frac{2 \text{MSE}}{b}}.$$

Tukey HSD (controlling the family-wise error rate):

$$\bar{y}_{i\bullet} - \bar{y}_{i'\bullet} \pm q_{g, (g-1)(b-1), \alpha} \cdot \sqrt{\frac{\text{MSE}}{b}},$$

where q is a quantile of the **Studentized range distribution**. The error degrees of freedom are $(g-1)(b-1)$ — the *correct* RCBD error d.f.

Remark

Contrasts, HSD, LSD... are computed exactly as for additive factorial data; one only needs the correct RCBD MSE and degrees of freedom.

Model Diagnostics

Always check the model assumptions

Four assumptions to inspect (as in any general linear model):

Linearity/additivity the residuals-vs-fitted plot shows *no* structure.

Constant variance (homoscedasticity) residual spread is even across fitted values.

Normality the Q–Q plot of the (studentized) residuals is near a straight line.

Independence ensured by randomization within blocks.

- ▶ **Box–Cox** suggests a response transformation if needed (e.g. $\lambda = 1/2$: square root).
- ▶ For the Air Freshener data: the residual and Q–Q plots are “fine”; Box–Cox suggests $\lambda \approx 1/2$ but $\lambda = 1$ (no transformation) is near the upper 95% CI limit, so it is acceptable.

Remark

Diagnose *before* interpreting p -values. A “significant” conclusion from a model with violated assumptions is not to be trusted.

Illustration with R

Example 1 — Read data & one-way analysis

```
# Read long-format data: store week trtmt sales
freshener <- read.table("air.freshener.txt", header = TRUE)
freshener$trtmt <- factor(freshener$trtmt) # A, B, C, D
freshener$store <- factor(freshener$store) # 8 blocks

# Analyse as a one-way CRD (ignore blocks)
lm0 <- lm(sales ~ trtmt, data = freshener)
anova(lm0)
```

```
Response: sales
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
trtmt	3	329	109.8	0.984	0.414
Residuals	28	3123	111.5		

⇒ ignoring blocks: treatment is **not significant** ($p = 0.41$).

Example 1 — The RCBD model (blocks included)

```
lm1 <- lm(sales ~ store + trtmt, data = freshener)
anova(lm1)
```

Response: sales

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
store	7	2478.9	354.1	11.536	0.0000059
trtmt	3	329.4	109.8	3.577	0.0312
Residuals	21	644.6	30.7		

- ▶ MSE drops sharply: $111.5 \rightarrow 30.7$ (block variation has been pulled out).
- ▶ Treatment is now **significant**: $F = 3.58$, $p = 0.031$.
- ▶ $SS_{\text{trt}} = 329.4$ is *unchanged* between the two models (balanced, orthogonal).
- ▶ Check: $644.6 + 2478.9 = 3123.5 = SSE_{\text{CRD}}$. ✓

Example 1 — Relative efficiency & multiple comparisons

```
MSblk ← 2478.9/7;  MSE ← 644.6/21;  g ← 4;  b ← 8
s2_crd ← ((b-1)*MSblk + b*(g-1)*MSE)/(g*b-1)
nu_r ← (g-1)*(b-1);  nu_c ← g*(b-1)
RE ← ((nu_r+1)*(nu_c+3))/((nu_r+3)*(nu_c+1)) * s2_crd/MSE
RE
# ~ 3.3

# Tukey HSD multiple comparisons for treatments
TukeyHSD(aov(sales ~ store + trtmt, data = freshener), "trtmt")

      diff      lwr      upr    p adj
B-A   1.75   -5.97    9.47  0.9206
D-A  -6.75  -14.47    0.97  0.1006
D-B  -8.50  -16.22   -0.78  0.0276  # ← significant
...    (D is lowest, the clearest difference)
```

Example 2 — Auditor Training (RCBD)

```
auditor <- read.table("AuditorTraining.txt", header = TRUE)
auditor$block      <- factor(auditor$block)          # 10 blocks
auditor$treatment  <- factor(auditor$treatment)      # 3 treatments
anova(lm(y ~ block + treatment, data = auditor))
```

Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
block	9	433.4	48.2	7.716	0.000132
treatment	2	1295.0	647.5	103.754	1.32e-10
Residuals	18	112.3	6.2		

⇒ the treatment difference is **very strong** ($F = 103.8$).

Example 2 — What if we drop the block?

```
anova(lm(y ~ treatment, data = auditor))
```

Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
treatment	2	1295.0	647.5	32.04	7.44e-08
Residuals	27	545.7	20.2		

- ▶ Still significant, but F falls from 103.8 $\rightarrow$ 32.0; MSE rises 6.2 $\rightarrow$ 20.2.
- ▶ $SSE_{CRD} = 545.7 = 112.3 + 433.4 = SSE_{RCBD} + SS_{block}$. ✓
- ▶ $RE \approx 3.0$: blocking on time of graduation is very effective.

Remark

Here dropping the block still “rescues” the conclusion (the signal is very strong), but for the Air Freshener data dropping the block *destroyed* the conclusion. Lesson: **do not** throw away block information.

Model diagnostics in R

```
lm1 <- lm(sales ~ store + trtmt, data = freshener)
par(mfrow = c(1,3))
plot(lm1$fitted, lm1$res, xlab="Fitted", ylab="Residuals"); abline(h=0)
qqnorm(rstudent(lm1)); qqline(rstudent(lm1))
library(MASS); boxcox(lm1) # suggest a transformation

# Tukey 1-df test for nonadditivity
z <- lm1$fitted^2 / (2*mean(freshener$sales))
anova(lm(sales ~ store + trtmt + z, data = freshener))
```

- ▶ Q-Q and residual plots are “fine”; Box-Cox suggests $\lambda \approx 1/2$, with $\lambda = 1$ acceptable.
- ▶ The z line checks nonadditivity: here $p \approx 0.065 > 0.05$, so we *do not* reject additivity at the 5% level (though it is close to the threshold — treat with care).

A Hand-Computation Example

A small data set to compute by hand ($g = 3$, $b = 4$)

A small experiment: 3 treatments, 4 blocks, one observation per cell.

Treatment	Block 1	Block 2	Block 3	Block 4	$\bar{y}_{i\bullet}$
A	7	7	8	10	8
B	7	11	10	12	10
C	10	12	12	14	12
$\bar{y}_{\bullet j}$	8	10	10	12	$\bar{y}_{\bullet\bullet} = 10$

- Treatment means: $\bar{y}_{A\bullet} = 8$, $\bar{y}_{B\bullet} = 10$, $\bar{y}_{C\bullet} = 12$.
- Block means: 8, 10, 10, 12. Grand mean: $\bar{y}_{\bullet\bullet} = 10$.

Steps 1–2: SS_{trt} and SS_{block}

Treatment sum of squares ($b = 4$):

$$SS_{\text{trt}} = b \sum_i (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})^2 = 4[(8 - 10)^2 + (10 - 10)^2 + (12 - 10)^2] = 4(4 + 0 + 4) = 32.$$

Block sum of squares ($g = 3$):

$$SS_{\text{block}} = g \sum_j (\bar{y}_{\bullet j} - \bar{y}_{\bullet\bullet})^2 = 3[(8 - 10)^2 + (10 - 10)^2 + (10 - 10)^2 + (12 - 10)^2] = 3(4 + 0 + 0 + 4) = 24.$$

Computation tip

Always compute SS_{trt} , SS_{block} from the *marginal means* then multiply by the factor (b or g).
You only need SST to obtain SSE by subtraction.

Steps 3–4: SST, SSE, and the ANOVA table

Total sum of squares:

$$SST = \sum_{i,j} (y_{ij} - \bar{y}_{..})^2 = 22 + 14 + 24 = 60$$

(by row: $A = 9+9+4+0 = 22$; $B = 9+1+0+4 = 14$; $C = 0+4+4+16 = 24$).

Error by subtraction: $SSE = SST - SS_{\text{trt}} - SS_{\text{block}} = 60 - 32 - 24 = 4$.

Source	d.f.	SS	MS	<i>F</i>
Block	3	24	8.00	—
Treatment	2	32	16.00	24.0
Error	6	4	0.667	
Total	11	60		

Conclusion of the hand example

$$F_{\text{trt}} = \frac{\text{MS}_{\text{trt}}}{\text{MSE}} = \frac{16.00}{0.667} = 24.0 \sim F_{2,6} \text{ under } H_0.$$

From tables: $F_{0.05; 2,6} = 5.14$. Since $24.0 > 5.14$ (in fact $p \approx 0.0014$), **reject H_0** : the three treatments differ.

Remark

If we *drop the block* and analyse as a CRD: $\text{SSE}_{\text{CRD}} = \text{SSE} + \text{SS}_{\text{block}} = 4 + 24 = 28$ with 9 degrees of freedom, $\text{MSE}_{\text{CRD}} = 3.11$. Then $F = 16/3.11 = 5.14$ — barely at the threshold, a much weaker conclusion. Blocking raised F from 5.1 to 24!

In-class exercise: compute $\hat{\alpha}_i, \hat{\beta}_j$ and the residual matrix e_{ij} yourself, and check that $\sum_i \hat{\alpha}_i = \sum_j \hat{\beta}_j = 0$ and $\sum_{i,j} e_{ij}^2 = 4$.

Extensions & Practical Notes

Missing data & imbalance

- ▶ A single missing value $\Rightarrow$ the design becomes **unbalanced**, and the marginal formulas $\bar{y}_{i\bullet}, \bar{y}_{\bullet j}$ no longer give orthogonal estimates.
- ▶ Then the **order** in which terms enter affects the sums of squares: use **Type II / III SS**, not sequential (Type I) SS.
- ▶ In R: `drop1(lm1, test="F")` or `car::Anova(lm1, type=2)`.
- ▶ One may impute the missing value, then subtract 1 error degree of freedom for each imputed cell.

Caution

With balanced data all three SS types coincide, so `anova(lm)` (Type I) is enough. Only under imbalance must you be careful.

Related block designs

Generalized RCB (GCBD) each treatment replicated $r > 1$ times per block $\Rightarrow$ the block $\times$ treatment interaction becomes estimable.

Latin square block in **two** directions at once (rows & columns), e.g. by both time and position.

Incomplete block when block size $<$ number of treatments (BIBD, PBIBD, lattice...).

Factorial in blocks (RCBF) the treatments have a factorial structure, set inside blocks.

Remark

The RCBD is the foundational “building brick”. Master it and the more complex block designs are just natural extensions of the same idea: *control noise by grouping similar units*.

Summary

The core things to remember

1. **Block** when you know of a large source of variation in advance: “block what you can”.
2. Model: $y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}$, **additive**, over-parameterized $\Rightarrow$ needs **2 constraints** ($\sum \alpha_i = \sum \beta_j = 0$).
3. Marginal estimates: $\hat{\mu} = \bar{y}_{\bullet\bullet}$, $\hat{\alpha}_i = \bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet}$, $\hat{\beta}_j = \bar{y}_{\bullet j} - \bar{y}_{\bullet\bullet}$.
4. Orthogonal decomposition: $SST = SS_{\text{trt}} + SS_{\text{block}} + SSE$; error d.f. = $(g - 1)(b - 1)$.
5. Treatment test: $F = MS_{\text{trt}}/MSE$. **Do not** test the block.
6. Good blocking $\Rightarrow$ small MSE, large **RE**, high power.
7. Check **additivity** (Tukey), **normality** (Q-Q), **variance** (residual plot).

Exercises

1. Prove $\sum_j \hat{\beta}_j = 0$ with $\hat{\beta}_j = \bar{y}_{\bullet j} - \bar{y}_{\bullet\bullet}$.
2. Prove in detail that the B–E cross term in the decomposition of SST equals 0.
3. Starting from $e_{ij} = \varepsilon_{ij} - \bar{\varepsilon}_{i\bullet} - \bar{\varepsilon}_{\bullet j} + \bar{\varepsilon}_{\bullet\bullet}$, compute $\text{Var}(e_{ij})$ and explain why $\mathbb{E}(\text{SSE}) = (g - 1)(b - 1)\sigma^2$.
4. Under reference coding ($\alpha_1 = \beta_1 = 0$), rewrite $\mathbb{E}(y_{ij})$ and show that the contrast $\alpha_2 - \alpha_3$ coincides with the result under the zero-sum constraint.

Practical exercises

1. For the hand-computation data ($g = 3, b = 4$): compute $\hat{\alpha}_i, \hat{\beta}_j$, the residual matrix, and rebuild the ANOVA table. Cross-check against R.
2. The AuditorTraining.txt data: fit the RCBD, compute RE, run Tukey HSD, draw diagnostic plots, and run the Tukey nonadditivity test.
3. The barley data, 5 varieties $\times$ several locations: analyse as an RCBD, compute RE, and estimate the number of CRD runs needed for the same power.
4. Simulation: generate RCBD data with large/small block effects and compare the power of the RCBD analysis with the CRD over 1000 replications.